Chapter 5: Advanced SQL

Database System Concepts, 7th Ed.

Outline

- Accessing SQL From a Programming Language
- Functions and Procedures
- Triggers

Accessing SQL from a Programming Language

A database programmer must have access to a general-purpose programming language for at least two reasons

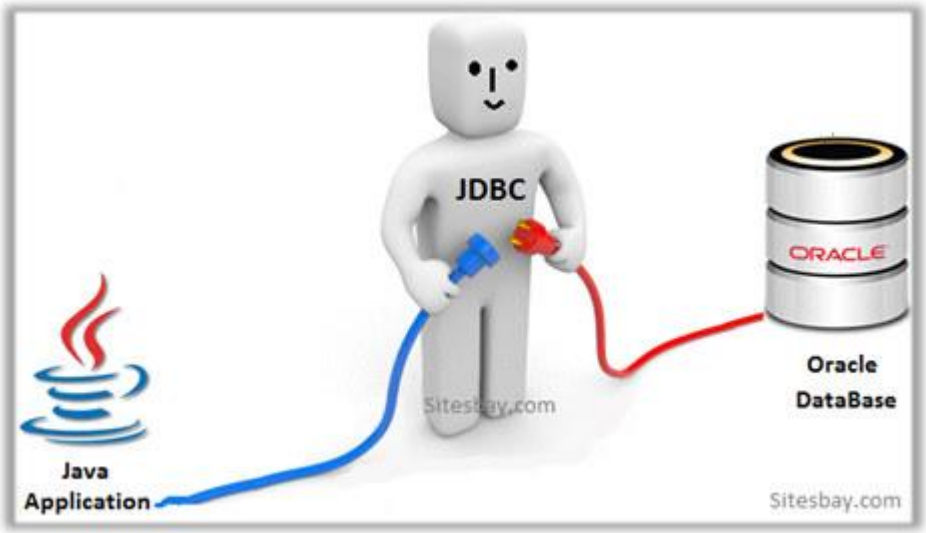
- Not all queries can be expressed in SQL, since SQL does not provide the full expressive power of a general-purpose language.
- Non-declarative actions -- such as printing a report, interacting with a user, or sending the results of a query to a graphical user interface -- cannot be done from within SQL.

Accessing SQL from a Programming Language (Cont.)

There are two approaches to accessing SQL from a general-purpose programming language

- Dynamic SQL -- A general-purpose program can connect to and communicate with a database server using a collection of functions
- Embedded SQL -- provides a means by which a program can interact with a database server.
  - The SQL statements are translated at compile time into function calls.
  - At runtime, these function calls connect to the database using an API that provides dynamic SQL facilities.

JDBC



https://learn.microsoft.com/zh-cn/sql/connect/jdbc/microsoft-jdbc-driver-for-sql-server?view=sql-server-ver16

JDBC

- **JDBC** is a Java API for communicating with database systems supporting SQL.
- JDBC supports a variety of features for querying and updating data, and for retrieving query results.
- Why JDBC?
  - Write once, run anywhere
    - Multiple client and server platforms.
  - Object-relational mapping
    - Databases optimized for searching/indexing.
    - Objects optimized for engineering/flexibility.
  - Network independence
    - Works across Internet Protocol
  - Database independence

JDBC

- JDBC also supports metadata retrieval, such as querying about relations present in the database and the names and types of relation attributes.
- Model for communicating with the database:
  - Open a connection
  - Create a "statement" object
  - Execute queries using the statement object to send queries and fetch results
  - Exception mechanism to handle errors

Schema of a JDBC Application

- Load the driver for a specific DBMS

  e.g., Class.forName("com.microsoft.jdbc.sqlserver.SQLServerDriver")
- Establish a connection to a specific database
- Create an abstract statement, to be sent over the connection
- Execute the statement by sending a Java string
  - E.g, returns an object of class ResultSet
- Process the result set with methods of ResultSet
- Close statement and connection

JDBC Code

```
public static void JDBCexample(String dbid, String userid, String passwd)
    {
     try (Connection conn = DriverManager.getConnection(
           "jdbc:oracle:thin:@db.yale.edu:2000:univdb", userid, passwd);
           Statement stmt = conn.createStatement();
         )
     {
           … Do Actual Work ….
     }
    catch (SQLException sqle) {
       System.out.println("SQLException : " + sqle);
     }
}
```

NOTE: Above syntax works with Java 7, and JDBC 4 onwards.
Resources opened in "try (….)" syntax ("try with resources") are automatically closed at the end of the try block

JDBC Code for Older Versions of Java/JDBC

```
public static void JDBCexample(String dbid, String userid, String passwd)
 {
   try {
       Class.forName ("oracle.jdbc.driver.OracleDriver");
       Connection conn = DriverManager.getConnection(
              "jdbc:oracle:thin:@db.yale.edu:2000:univdb", userid, passwd);
      Statement stmt = conn.createStatement();
         … Do Actual Work ….
      stmt.close();
      conn.close();
   }
   catch (SQLException sqle) {
      System.out.println("SQLException : " + sqle);
   }
}
```

NOTE: Class.forName is not required from JDBC 4 onwards. The try with resources syntax in prev slide is preferred for Java 7 onwards.

JDBC Code (Cont.)

- Update to database

```
try {
    stmt.executeUpdate(
        "insert into instructor values('77987', 'Kim', 'Physics', 98000)");
} catch (SQLException sqle)
{
   System.out.println("Could not insert tuple. " + sqle);
}
```

- Execute query and fetch and print results

```
ResultSet rset = stmt.executeQuery(
                          "select dept_name, avg (salary)
                           from instructor
                           group by dept_name");
while (rset.next()) {
      System.out.println(rset.getString("dept_name") + " " +
                                    rset.getFloat(2));
}
```

JDBC SUBSECTIONS

- Connecting to the Database
- Shipping SQL Statements to the Database System
- Exceptions and Resource Management
- Retrieving the Result of a Query
- Prepared Statements
- Callable Statements
- Metadata Features
- Other Features
- Database Access from Python

JDBC Code Details

- Getting result fields:
  - **rs.getString("dept_name") and rs.getString(1) equivalent if dept_name is the first argument of select result.**
- Dealing with Null values

```
int a = rs.getInt( "a" );
if (rs.wasNull()) Systems.out.println( "Got null value" );
```

SQL Injection

- Suppose query is constructed using
  - "select * from instructor where name = '" + name + "'"
- Suppose the user, instead of entering a name, enters:
  - X' or 'Y' = 'Y
- then the resulting statement becomes:
  - "select * from instructor where name = '" + "X' or 'Y' = 'Y" + "'"
  - which is:
    - select * from instructor where name = 'X' or 'Y' = 'Y'
  - User could have even used
    - X'; update instructor set salary = salary + 10000; --
- Prepared stament internally uses:
  "select * from instructor where name = 'X\' or \'Y\' = \'Y'
  - **Always use prepared statements, with user inputs as parameters**

Prepared Statement

- PreparedStatement pStmt = conn.prepareStatement(
  "insert into instructor values(?,?,?,?)");
- pStmt.setString(1, "88877");
  pStmt.setString(2, "Perry");
  pStmt.setString(3, "Finance");
  pStmt.setInt(4, 125000);
  pStmt.executeUpdate();
  pStmt.setString(1, "88878");
  pStmt.executeUpdate();
- WARNING: always use prepared statements when taking an input from the user and adding it to a query
  - NEVER create a query by concatenating strings
  - "insert into instructor values(' " + ID + " ', ' " + name + " ', " + " ' + dept name + " ', " ' balance + ') “
  - What if name is “D'Souza” ?

Prepared Statement

- String sql = "SELECT * FROM users WHERE username = ? AND password = ?";
- PreparedStatement stmt = conn.prepareStatement(sql);
- String userInputUsername = "admin";
- String userInputPassword = "password123";

// 设置查询参数

- stmt.setString(1, userInputUsername);
- stmt.setString(2, userInputPassword);pStmt.executeUpdate();

Metadata Features

- ResultSet metadata
- E.g.after executing query to get a ResultSet rs:
  - ResultSetMetaData rsmd = rs.getMetaData();

```
for(int i = 1; i <= rsmd.getColumnCount(); i++) {
        System.out.println(rsmd.getColumnName(i));
        System.out.println(rsmd.getColumnTypeName(i));
}
```

- How is this useful?

Metadata (Cont)

- Database metadata
- DatabaseMetaData dbmd = conn.getMetaData();

```
// Arguments to getColumns: Catalog, Schema-pattern, Table-pattern,
// and Column-Pattern
// Returns: One row for each column; row has a number of attributes
// such as COLUMN_NAME, TYPE_NAME
// The value null indicates all Catalogs/Schemas.
// The value “” indicates current catalog/schema
// The value “%” has the same meaning as SQL like clause
ResultSet rs = dbmd.getColumns(null, "univdb", "department", "%");
while( rs.next()) {
        System.out.println(rs.getString("COLUMN_NAME"),
                                rs.getString("TYPE_NAME");
}
```

- And where is this useful?

Metadata (Cont)

- Database metadata
- DatabaseMetaData dbmd = conn.getMetaData();

```
// Arguments to getTables: Catalog, Schema-pattern, Table-pattern,
// and Table-Type
// Returns: One row for each table; row has a number of attributes
// such as TABLE_NAME, TABLE_CAT, TABLE_TYPE, ..
// The value null indicates all Catalogs/Schemas.
// The value “” indicates current catalog/schema
// The value “%” has the same meaning as SQL like clause
// The last attribute is an array of types of tables to return.
//    TABLE means only regular tables
ResultSet rs = dbmd.getTables (“”, "", “%", new String[] {“TABLES”});
while( rs.next()) {
        System.out.println(rs.getString(“TABLE_NAME“));
}
```

- And where is this useful?

Finding Primary Keys

- DatabaseMetaData dmd = connection.getMetaData();

```
// Arguments below are:  Catalog, Schema, and Table
// The value “”  for Catalog/Schema indicates current catalog/schema
//  The value null indicates all catalogs/schemas
ResultSet rs = dmd.getPrimaryKeys(“”, “”, tableName);

while(rs.next()){
    // KEY_SEQ indicates the position of the attribute in
    // the primary key, which is required if a primary key has multiple
    // attributes
    System.out.println(rs.getString(“KEY_SEQ”),
                                rs.getString("COLUMN_NAME");
}
```

Transaction Control in JDBC

- By default, each SQL statement is treated as a separate transaction that is committed automatically
  - bad idea for transactions with multiple updates
- Can turn off automatic commit on a connection
  - conn.setAutoCommit(false);
- Transactions must then be committed or rolled back explicitly
  - conn.commit(); or
  - conn.rollback();
- conn.setAutoCommit(true) turns on automatic commit.

Other JDBC Features

- Calling functions and procedures
  - CallableStatement cStmt1 = conn.prepareCall("{? = call some function(?)}");
  - CallableStatement cStmt2 = conn.prepareCall("{call some procedure(?,?)}");
- Handling large object types
  - getBlob() and getClob() that are similar to the getString() method, but return objects of type Blob and Clob, respectively
  - get data from these objects by getBytes()
  - associate an open stream with Java Blob or Clob object to update large objects
    - blob.setBlob(int parameterIndex, InputStream inputStream).

SQLJ

- JDBC is overly dynamic, errors cannot be caught by compiler
- SQLJ: embedded SQL in Java
  - #sql iterator deptInfoIter (String dept name, int avgSal);

```
deptInfoIter iter = null;
#sql iter = { select dept_name, avg(salary) from instructor
                group by dept name };
while (iter.next()) {
    String deptName = iter.dept_name();
     int avgSal = iter.avgSal();
     System.out.println(deptName + " " + avgSal);
}
iter.close();
```

ODBC

ODBC

- Open DataBase Connectivity (ODBC) standard
  - standard for application program to communicate with a database server.
  - application program interface (API) to
    - open a connection with a database,
    - send queries and updates,
    - get back results.
- Applications such as GUI, spreadsheets, etc. can use ODBC
- https://learn.microsoft.com/zh-cn/sql/connect/odbc/microsoft-odbc-driver-for-sql-server?view=sql-server-ver16

Embedded SQL

- The SQL standard defines embeddings of SQL in a variety of programming languages such as C, C++, Java, Fortran, and PL/1,
- A language to which SQL queries are embedded is referred to as a **host language**, and the SQL structures permitted in the host language comprise *embedded* SQL.
- The basic form of these languages follows that of the System R embedding of SQL into PL/1.
- **EXEC SQL** statement is used in the host language to identify embedded SQL request to the preprocessor

  EXEC SQL <embedded SQL statement >;

  Note: this varies by language:
  - In some languages, like COBOL, the semicolon is replaced with END-EXEC
  - In Java embedding uses # SQL { …. };

Embedded SQL (Cont.)

- Before executing any SQL statements, the program must first connect to the database. This is done using:

  EXEC-SQL connect to server user user-name using password;

  Here, *server* identifies the server to which a connection is to be established.

- Variables of the host language can be used within embedded SQL statements. They are preceded by a colon (:) to distinguish from SQL variables (e.g., :*credit_amount*)
- Variables used as above must be declared within DECLARE section, as illustrated below. The syntax for declaring the variables, however, follows the usual host language syntax.

```
EXEC-SQL BEGIN DECLARE SECTION;
    int credit-amount ;
EXEC-SQL END DECLARE SECTION;
```

Embedded SQL (Cont.)

- To write an embedded SQL query, we use the

  declare *c* **cursor for** **<SQL query>**

  statement. The variable *c* is used to identify the query
- Example:
  - From within a host language, find the ID and name of students who have completed more than the number of credits stored in variable credit_amount in the host langue
  - Specify the query in SQL as follows:

```
EXEC SQL
    declare c cursor for
    select ID, name
    from student
    where tot_cred > :credit_amount
END_EXEC
```

Embedded SQL (Cont.)

- The **open** statement for our example is as follows:

  EXEC SQL **open** *c* ;

  This statement causes the database system to execute the query and to save the results within a temporary relation. The query uses the value of the host-language variable *credit-amount* at the time the **open** statement is executed.

- The fetch statement causes the values of one tuple in the query result to be placed on host language variables.

  EXEC SQL **fetch** *c* **into** :*si, :sn* END_EXEC

  Repeated calls to fetch get successive tuples in the query result

Embedded SQL (Cont.)

- A variable called SQLSTATE in the SQL communication area (SQLCA) gets set to '02000' to indicate no more data is available
- The **close** statement causes the database system to delete the temporary relation that holds the result of the query.

  EXEC SQL **close** *c* ;

  Note: above details vary with language. For example, the Java embedding defines Java iterators to step through result tuples.

Updates Through Embedded SQL

- Embedded SQL expressions for database modification (**update**, **insert**, and **delete**)
- Can update tuples fetched by cursor by declaring that the cursor is for update

```
EXEC SQL
  declare c cursor for
    select *
    from instructor
    where dept_name = 'Music'
    for update
```

- We then iterate through the tuples by performing **fetch** operations on the cursor (as illustrated earlier), and after fetching each tuple we execute the following code:

```
update instructor
set salary = salary + 1000
where current of c
```

Embedded SQL Example

- https://learn.microsoft.com/en-us/sql/odbc/reference/embedded-sql-example?view=sql-server-ver16

Functions and Procedures

Functions and Procedures

- Functions and procedures allow "business logic" to be stored in the database and executed from SQL statements.
- These can be defined either by the procedural component of SQL or by an external programming language such as Java, C, or C++.
- The syntax we present here is defined by the SQL standard.
  - Most databases implement nonstandard versions of this syntax.

Declaring SQL Functions

- Define a function that, given the name of a department, returns the count of the number of instructors in that department.

```
create function dept_count (dept_name varchar(20))
        returns integer
        begin
        declare d_count  integer;
              select count (* ) into d_count
              from instructor
              where instructor.dept_name = dept_name
        return d_count;
end
```

- The function *dept*_count can be used to find the department names and budget of all departments with more that 12 instructors.

```
select dept_name, budget
from department
where dept_count (dept_name ) > 12
```

https://learn.microsoft.com/zh-cn/sql/relational-databases/user-defined-functions/user-defined-functions?view=sql-server-ver16

Table Functions

- The SQL standard supports functions that can return tables as results; such functions are called **table functions**
- Example: Return all instructors in a given department

```
create function instructor_of (dept_name char(20))
        returns table  (
                ID varchar(5),
                name varchar(20),
                dept_name varchar(20),
                salary numeric(8,2))
        return table
                (select ID, name, dept_name, salary
                from instructor
                where instructor.dept_name = instructor_of.dept_name)
```

- Usage

```
select *
from table (instructor_of ('Music'))
```

SQL Procedures

- The *dept_count* function could instead be written as procedure:

```
create procedure dept_count_proc (in dept_name varchar(20),
                                  out d_count integer)
  begin
     select count(*) into d_count
     from instructor
     where instructor.dept_name = dept_count_proc.dept_name
  end
```

- The keywords **in** and **out** are parameters that are expected to have values assigned to them and parameters whose values are set in the procedure in order to return results.
- Procedures can be invoked either from an SQL procedure or from embedded SQL, using the **call** statement.

```
declare d_count integer;
call dept_count_proc( 'Physics', d_count);
```

https://learn.microsoft.com/zh-cn/sql/relational-databases/stored-procedures/stored-procedures-database-engine?view=sql-server-ver16

SQL Procedures (Cont.)

- Procedures and functions can be invoked also from dynamic SQL
- SQL allows more than one procedure of the so long as the number of arguments of the procedures with the same name is different.
- The name, along with the number of arguments, is used to identify the procedure.

Language Constructs for Procedures & Functions

- SQL supports constructs that gives it almost all the power of a general-purpose programming language.
  - Warning: most database systems implement their own variant of the standard syntax below.
- Compound statement: **begin** … **end,**
  - May contain multiple SQL statements between **begin** and **end**.
  - Local variables can be declared within a compound statements
- While and repeat statements:
  - **while** boolean expression **do**
    sequence of statements ;
    end while
  - **repeat**
    sequence of statements ;
    until boolean expression
    end repeat

Language Constructs (Cont.)

- **For** loop
  - Permits iteration over all results of a query
- Example: Find the budget of all departments

```
declare n integer default 0;
for r as
        select budget from department
       where dept_name = 'Music'
 do
              set n = n + r.budget
 end for
```

Language Constructs – if-then-else

- Conditional statements (**if-then-else**)

if *boolean expression*
then *statement or compound statement*
elseif *boolean expression*
then *statement or compound statement*
else *statement or compound statement*
end if

Example procedure

- Registers student after ensuring classroom capacity is not exceeded
  - Returns 0 on success and -1 if capacity is exceeded
  - See book (page 202) for details
- Signaling of exception conditions, and declaring handlers for exceptions

  declare *out_of_classroom_seats* **condition**
  declare exit handler for *out_of_classroom_seats*
  begin
  …
  end
- The statements between the **begin** and the **end** can raise an exception by executing "**signal** *out_of_classroom_seats*"
- The handler says that if the condition arises he action to be taken is to exit the enclosing the **begin end** statement.

练习：写一个函数找出所有的supervisors

- Relation *emp-super*

person	*supervisor*
Bob	Alice
Mary	Susan
Alice	David
David	Mary

- Find the supervisor of “Bob”
- Find the supervisor of the supervisor of “Bob”
- Can you find ALL the supervisors (direct and indirect) of “Bob”?

External Language Routines

- SQL allows us to define functions in a programming language such as Java, C#, C or C++.
  - Can be more efficient than functions defined in SQL, and computations that cannot be carried out in SQL\can be executed by these functions.
- Declaring external language procedures and functions

```
create procedure dept_count_proc(in dept_name varchar(20),
                                 out count integer)
language C
external name   '/usr/avi/bin/dept_count_proc'

create function dept_count(dept_name varchar(20))
returns integer
language C
external name '/usr/avi/bin/dept_count'
```

External Language Routines (Cont.)

- Benefits of external language functions/procedures:
  - more efficient for many operations, and more expressive power.
- Drawbacks
  - Code to implement function may need to be loaded into database system and executed in the database system's address space.
    - risk of accidental corruption of database structures
    - security risk, allowing users access to unauthorized data
  - There are alternatives, which give good security at the cost of potentially worse performance.
  - Direct execution in the database system' s space is used when efficiency is more important than security.

Security with External Language Routines

- To deal with security problems, we can do on of the following:
  - Use **sandbox** techniques
    - That is, use a safe language like Java, which cannot be used to access/damage other parts of the database code.
  - Run external language functions/procedures in a separate process, with no access to the database process' memory.
    - Parameters and results communicated via inter-process communication
- Both have performance overheads
- Many database systems support both above approaches as well as direct executing in database system address space.

Triggers

https://learn.microsoft.com/zh-cn/sql/t-sql/statements/create-trigger-transact-sql?view=sql-server-ver17

Triggers

- A **trigger** is a statement that is executed automatically by the system as a side effect of a modification to the database.
- To design a trigger mechanism, we must:
  - Specify the conditions under which the trigger is to be executed.
  - Specify the actions to be taken when the trigger executes.
- Triggers introduced to SQL standard in SQL:1999, but supported even earlier using non-standard syntax by most databases.
  - Syntax illustrated here may not work exactly on your database system; check the system manuals

Triggering Events and Actions in SQL

- Triggering event can be **insert**, **delete** or **update**
- Triggers on update can be restricted to specific attributes
  - For example, **after update of** *takes* **on** *grade*
- Values of attributes before and after an update can be referenced
  - **referencing old row as** : for deletes and updates
  - **referencing new row as** : for inserts and updates
- Triggers can be activated before an event, which can serve as extra constraints. For example, convert blank grades to null.

```
create trigger setnull_trigger before update of takes
referencing new row as nrow
for each row
      when (nrow.grade = ' ')
       begin atomic
          set nrow.grade = null;
 end;
```

Trigger to Maintain credits_earned value

- **create trigger** *credits_earned* **after update of** *takes* **on** (*grade*)
  referencing new row as *nrow*
  referencing old row as *orow*
  for each row
  when *nrow.grade* <> 'F' **and** *nrow.grade* **is not null**
  and (*orow.grade* = 'F' **or** *orow.grade* **is null**)
  begin atomic
  update *student*
  set *tot_cred*= *tot_cred* +
  (**select** *credits*
  from *course*
  where *course.course_id*= *nrow.course_id*)
  where *student.id* = *nrow.id*;
  end;

Statement Level Triggers

- Instead of executing a separate action for each affected row, a single action can be executed for all rows affected by a transaction
  - Use **for each statement** instead of **for each row**
  - Use **referencing old table** or **referencing new table** to refer to temporary tables (called ***transition tables***) containing the affected rows
  - Can be more efficient when dealing with SQL statements that update a large number of rows

When Not To Use Triggers

- Triggers were used earlier for tasks such as
  - Maintaining summary data (e.g., total salary of each department)
  - Replicating databases by recording changes to special relations (called **change** or **delta** relations) and having a separate process that applies the changes over to a replica
- There are better ways of doing these now:
  - Databases today provide built in materialized view facilities to maintain summary data
  - Databases provide built-in support for replication
- Encapsulation facilities can be used instead of triggers in many cases
  - Define methods to update fields
  - Carry out actions as part of the update methods instead of through a trigger

When Not To Use Triggers (Cont.)

- Risk of unintended execution of triggers, for example, when
  - Loading data from a backup copy
  - Replicating updates at a remote site
  - Trigger execution can be disabled before such actions.
- Other risks with triggers:
  - Error leading to failure of critical transactions that set off the trigger
  - Cascading execution

触发器概述

- 触发器的常用功能

(1) 完成比约束更复杂的数据约束。

(2) 触发器可以检查T-SQL所做的操作是否被允许。

例如：在产品库存表里，如果要对某种产品出库，触发器可以检查该产品最低库存数。

(3) 当一个T-SQL语句对数据表进行操作的时候，触发器可以根据该T-SQL语句的操作情况来对另一个数据表进行操作。

(4) 触发器也可以调用一个或多个存储过程。

(5) 在T-SQL语句执行完之后，触发器可自动调用“数据库邮件”来发送邮件。

触发器概述

- 触发器的分类：DML触发器
- DML触发器是当数据库服务器中发生数据操作语言事件时执行的存储过程。DML触发器又分为两类：
  - AFTER触发器：只有执行某一操作INSERT、UPDATE、DELETE之后触发器才被触发；
  - INSTEAD OF触发器：即当对表进行INSERT、UPDATE 或 DELETE 操作时，系统不是直接对表执行这些操作，而是把操作内容交给触发器，让触发器检查所进行的操作是否正确，如正确才进行相应的操作。因此，INSTEAD OF 触发器的动作要早于表约束的处理。

触发器概述

- Instead of触发器和After触发器区别
  - Instead of触发在执行表约束检查之前执行。除表之外，Instead of 触发器也可以用于视图，用来扩展视图可以支持的更新操作。
  - After触发器在一个Insert,Update或Deleted语句之后执行，进行约束检查等动作都在After触发器被激活之前发生。After触发器只能用于表。
  - INSTEAD OF 触发器的操作有点类似于完整性约束。在对数据库的操纵时，有些情况下使用约束可以达到更好的效果，而如果采用触发器，则能定义比完整性约束更加复杂的约束。

创建触发器

- DML触发器

```
CREATE TRIGGER <触发器名>
ON  <表名>|<视图名>
FOR | AFTER | INSTEAD OF
[INSERT][,UPDATE][,DELETE]
AS
    <T-SQL语句> | <语句块>
```

- FOR | AFTER | INSTEAD OF：指定触发器触发的时机
- AFTER指定在相应操作（INSERT、UPDATE、DELETE）成功执行后才触发。FOR=AFTER
- INSTEAD OF指定执行DML触发器用于“代替”引发触发器执行的INSERT、UPDATE或DELETE语句。

创建触发器

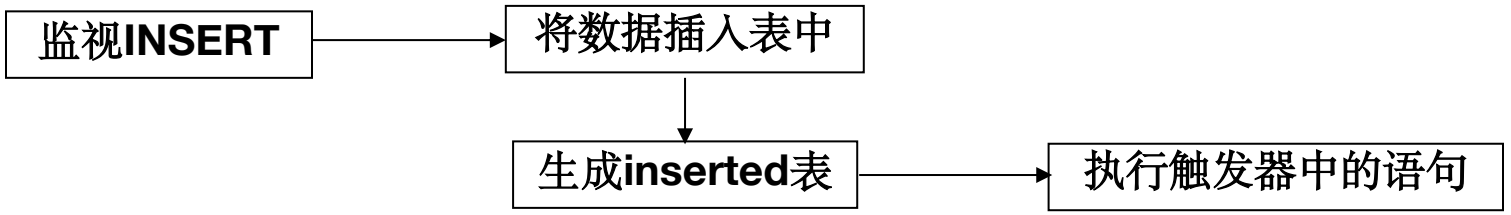
例 用T-SQL语句为S表创建一个DELETE类型的触发器DEL_COUNT，删除数据时，显示删除学生的个数。

```
CREATE TRIGGER DEL_COUNT
ON S
FOR DELETE
AS
    DECLARE @COUNT VARCHAR(50)
    SELECT @COUNT=STR(@@ROWCOUNT)+'个学生被删除'
    SELECT @COUNT
RETURN
```

触发器说明

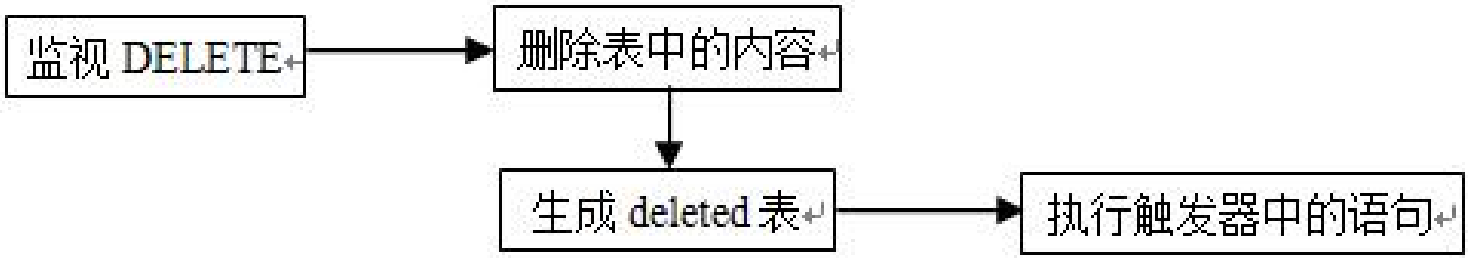
触发器中使用的特殊表

- 执行触发器时，系统创建了两个特殊的临时表inserted表和deleted表。
- inserted表：当向数据表中插入数据时，INSERT触发器触发执行，新元组插入到数据表和临时表inserted中。inserted表是一个逻辑表，它包含了已经插入的数据元组的一个副本。INSERT触发器的工作过程如图。



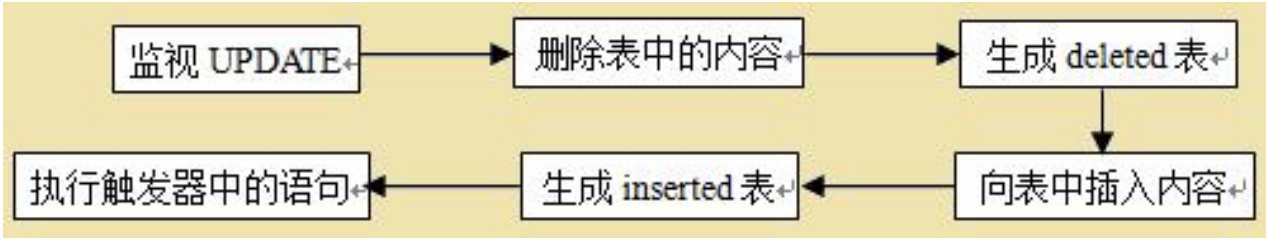
触发器说明

- deleted表：当试图删除表中元组时，DELETE触发器触发执行，被删除的元组存放到deleted表中。deleted表是一个逻辑表，它包含了已经删除数据元组的一个副本。DELETE触发器的工作过程如图。



触发器说明

- UPDATE触发器：当试图更新表中元组数据时，UPDATE触发器触发执行，UPDATE语句的执行可以看成两步，即先删除，后插入。因此执行过程中把数据表中原元组先移到deleted表中，再把修改过的元组插入到inserted表中。UPDATE触发器的工作过程如图所示。



触发器说明

- 由于inserted表和deleted表都是临时表，它们在触发器执行时被创建，触发器执行完后就消失了，所以只可以在触发器的语句中使用SELECT语句查询这两个表。
- 例：

```
--创建学生表
create table student(
    stu_id int identity(1,1) primary key,
    stu_name varchar(10),
    stu_gender char(2),
    stu_age int
)
----创建存储学生人数的student_sum表
create table student_sum(stuCount int default(0));
```

触发器说明

- 例：功能是向student插入数据的同时级联插入到student_sum表中，更新stuCount

```
--创建insert触发器
create trigger trig_insert
on student
after insert
as
begin
    declare @stuNumber int;
    select @stuNumber = count(*) from student;
    if not exists (select * from student_sum)--判断表中是否有记录
        insert into student_sum values(0);
    update student_sum set stuCount =@stuNumber; --把更新后总的学生数插入到student_sum表中
end
```

触发器说明

- 例:

```
--创建delete触发器
create trigger trig_delete
on student
after delete
as
begin
    select stu_id as 已删除的学生编号,stu_name stu_gender,stu_age
    from deleted
end;
```

触发器说明

- 例：

```
--创建update触发器
create trigger trig_update
on student
after update
as
begin
    declare @stuCount int;
    select @stuCount=count(*) from student;
    update student_sum set stuCount =@stuCount;
    select stu_id as 更新前学生编号,stu_name as 更新前学生姓名 from deleted
    select stu_id as 更新后学生编号,stu_name as 更新后学生姓名 from inserted
end
```

Recursive Queries

Recursion in SQL

- SQL:1999 permits recursive view definition
- Example: find which courses are a prerequisite, whether directly or indirectly, for a specific course

```
with recursive rec_prereq(course_id, prereq_id) as (
        select course_id, prereq_id
        from prereq
    union
        select rec_prereq.course_id, prereq.prereq_id,
        from rec_rereq, prereq
        where rec_prereq.prereq_id = prereq.course_id
    )
select *
from rec_prereq;
```

This example view, *rec_prereq,* is called the *transitive closure* of the *prereq* relation

The Power of Recursion

- Recursive views make it possible to write queries, such as transitive closure queries, that cannot be written without recursion or iteration.
  - Intuition: Without recursion, a non-recursive non-iterative program can perform only a fixed number of joins of *prereq* with itself
    - This can give only a fixed number of levels of managers
    - Given a fixed non-recursive query, we can construct a database with a greater number of levels of prerequisites on which the query will not work
    - Alternative: write a procedure to iterate as many times as required
      - See procedure *findAllPrereqs* in book

The Power of Recursion

- Computing transitive closure using iteration, adding successive tuples to *rec_prereq*
  - The next slide shows a *prereq* relation
  - Each step of the iterative process constructs an extended version of *rec_prereq* from its recursive definition.
  - The final result is called the *fixed point* of the recursive view definition.
- Recursive views are required to be **monotonic**. That is, if we add tuples to *prereq* the view *rec_prereq* contains all of the tuples it contained before, plus possibly more

Example of Fixed-Point Computation

course_id	*prereq_id*
BIO-301	BIO-101
BIO-399	BIO-101
CS-190	CS-101
CS-315	CS-190
CS-319	CS-101
CS-319	CS-315
CS-347	CS-319

Iteration Number	*Tuples in c1*
0	
1	(CS-319)
2	(CS-319), (CS-315), (CS-101)
3	(CS-319), (CS-315), (CS-101), (CS-190)
4	(CS-319), (CS-315), (CS-101), (CS-190)
5	done

Advanced Aggregation Features

Ranking

- Ranking is done in conjunction with an order by specification.
- Suppose we are given a relation
  student_grades(ID, GPA)
  giving the grade-point average of each student
- Find the rank of each student.
- **select** *ID*, **rank**() **over** (**order by** *GPA* **desc) as** *s_rank*
  from *student_grades*
- An extra **order by** clause is needed to get them in sorted order

  select *ID*, **rank**() **over** (**order by** *GPA* **desc) as** *s_rank*
  from *student_grades*
  order by *s_rank*
- Ranking may leave gaps: e.g. if 2 students have the same top GPA, both have rank 1, and the next rank is 3
  - **dense_rank** does not leave gaps, so next dense rank would be 2

Ranking

- Ranking can be done using basic SQL aggregation, but resultant query is very inefficient

```
select ID, (1 + (select count(*)
                 from student_grades B
                 where B.GPA > A.GPA)) as s_rank
from student_grades A
order by s_rank;
```

Ranking (Cont.)

- Ranking can be done within partition of the data.
- "Find the rank of students within each department."

```
select ID, dept_name,
        rank () over (partition by dept_name order by GPA desc)
                as dept_rank
from dept_grades
order by dept_name, dept_rank;
```

- Multiple **rank** clauses can occur in a single **select** clause.
- Ranking is done *after* applying **group by** clause/aggregation
- Can be used to find top-n results
  - More general than the **limit** *n* clause supported by many databases, since it allows top-n within each partition

Ranking (Cont.)

- Other ranking functions:
  - **percent_rank** (within partition, if partitioning is done)
  - **cume_dist** (cumulative distribution)
    - fraction of tuples with preceding values
  - **row_number** (non-deterministic in presence of duplicates)
- SQL:1999 permits the user to specify **nulls first** or **nulls last**

```
select ID,
        rank ( ) over (order by GPA desc nulls last) as s_rank
from student_grades
```

Ranking (Cont.)

- For a given constant *n*, the ranking the function *ntile*(*n*) takes the tuples in each partition in the specified order, and divides them into *n* buckets with equal numbers of tuples.
- E.g.,

  select *ID*, **ntile**(4) **over** (**order by** *GPA* **desc**) **as** *quartile*
  from *student_grades;*

Windowing

- Used to smooth out random variations.
- E.g., **moving average**: "Given sales values for each date, calculate for each date the average of the sales on that day, the previous day, and the next day"
- **Window specification** in SQL:
  - Given relation *sales(date, value)*

```
select date, sum(value) over
      (order by date between rows 1 preceding and 1 following)
 from sales
```

Windowing

- Examples of other window specifications:
  - **between rows unbounded preceding and current**
  - **rows unbounded preceding**
  - **range between** 10 **preceding and current row**
    - All rows with values between current row value –10 to current value
  - **range interval** 10 **day preceding**
    - Not including current row

Windowing (Cont.)

- Can do windowing within partitions
- E.g., Given a relation *transaction* (*account_number, date_time, value*), where value is positive for a deposit and negative for a withdrawal
  - "Find total balance of each account after each transaction on the account"

```
select account_number, date_time,
    sum (value) over
                (partition by account_number
                order by date_time
                rows unbounded preceding)
      as balance
from transaction
order by account_number, date_time
```

End of Chapter 5